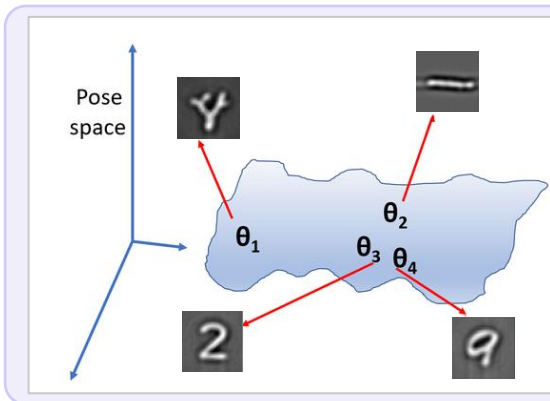


Cross-Harmonic MPI Restoration via Generative Models

Team Legend_intern

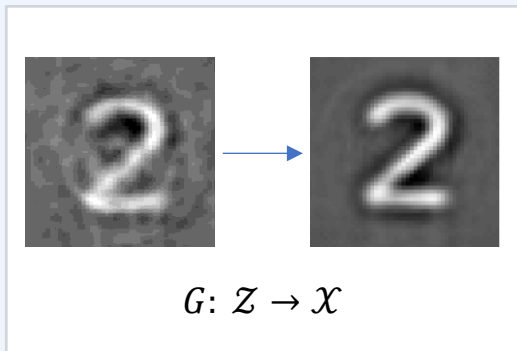
What is a Generative Model?

Learning the Data Manifold



- **Manifold Hypothesis:** High-dimensional medical images reside on lower-dimensional, continuous manifold.
- **Intrinsic Representation:** Learns latent parameters (shape, intensity) defining the data distribution instead of pixel-wise memorization.
- **Structural Logic:** Sampling from the manifold ensures "plausible" realizations and anatomical integrity.

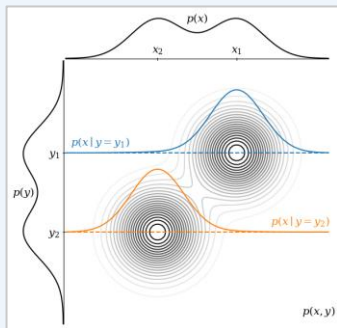
From Prior to Data



- **Mapping Function (G):** Transforms a simple prior ($z \sim p(z)$, e.g., Gaussian noise) into the complex target distribution $p_{\text{data}}(x)$.
$$x = G(z), \quad z \sim \mathcal{N}(0, \mathbf{I})$$
- **Data-Driven Prior:** Synthesizes missing high-frequency details, surpassing classical hand-crafted solvers.
- **Predictive Synthesis:** Statistically infers information physically lost to noise or signal decay.

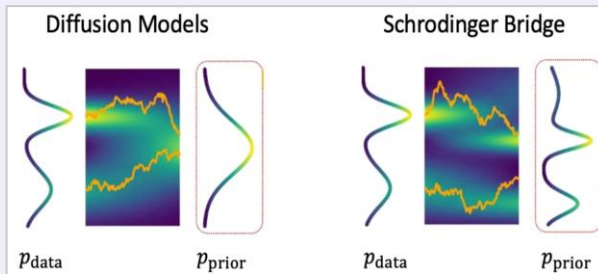
Generative Modeling as a Distribution Bridge

Conditional Probabilistic Mapping



- **Domain Translation:** Shift from "unconditional" creation to mapping between signal domains ($y \rightarrow x$)
- **Formulating $p(x|y)$:** Finding the most likely high-fidelity target (x) given a corrupted or low-resolution observation (y).
- **Statistical Bridge:** Connects stable, high-SNR inputs with sharp, high-resolution outputs.

The Concept of Trajectory



- **Continuous Path:** Restoration as a multi-step trajectory rather than a single-step operation.
- **Forward and Reverse Dynamics:**
 - **Forward ($x_0 \rightarrow x_1$):** Models degradation via noise or blurring.
 - **Reverse ($x_1 \rightarrow x_0$):** Inverts the path to recover manifold details.
- **Mathematical Flexibility:** Utilizes Stochastic Differential Equations (SDE) or Deterministic Flows for precise mapping.

Why Generative Models?

01 Distribution Learning

Learn $p(x)$ from data, not hand-crafted priors

Captures complex anatomical structures that traditional regularization cannot model

02 Missing Info Recovery

Fill in degraded details from learned prior

Generates plausible high-frequency content that was lost or corrupted in measurement

03 Uncertainty Awareness

Stochastic sampling captures reconstruction uncertainty

Unlike deterministic regression, provides multiple plausible solutions for ill-posed problems

Representative Success Cases in Biomedical Imaging

Accelerated MRI

Score-MRI

Chung & Ye, 2022

Medical Image Analysis

- Score-based diffusion as prior
- Unsupervised learning
- Beats supervised methods
- Any sampling pattern

Inverse Problem

DPS

Chung et al., 2023

ICLR 2023 (Top 25%)

- General noisy inverse solver
- Linear & nonlinear problems
- Gaussian / Poisson noise
- Posterior sampling

Low-dose CT

SAD

Du et al., 2024

Phys. Med. Biol.

- Nonlinear diffusion bridge
- Structure-aware prompts
- Clean $\leftrightarrow$ degraded distribution
- High-fidelity reconstruction

Cross-modality

SynDiff

Özbey et al., 2023

IEEE TMI

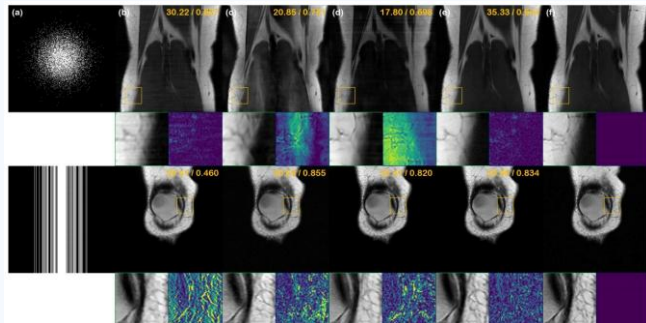
- Adversarial diffusion model
- MRI $\leftrightarrow$ CT translation
- Unpaired training (cycle)
- High-fidelity synthesis

These successes across CT, MRI, and inverse problems → **Can we apply the same strategy to MPI?**

Representative Results

Accelerated MRI

Score-MRI Chung & Ye, 2022



Multi-coil MRI reconstruction: (a) mask, (b) TV, (c) U-Net, (d) E2E-varnet, (e) Score-MRI, (f) GT

Low-dose CT

SAD Du et al., 2024

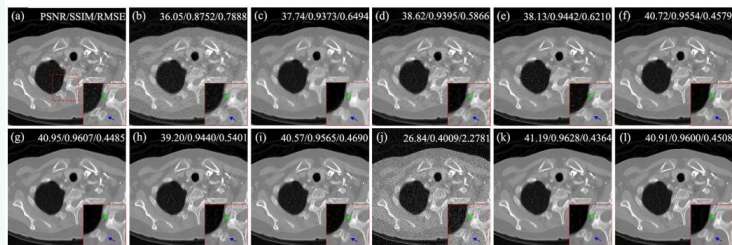


Figure 4. Qualitative results of different methods from the LoDoPaB-CT testing set. (a) Ground Truth, (b) FBP, (c) TV, (d) FBPNNet, (e) RED-CNN, (f) LearnPD, (g) CNCL-UNet, (h) iDDPM-1000, (i) iDDPM-50, (j) iDDPM-5, (k) SAD-5, (l) SAD-1. The display window is [-1001, 831]HU. The arrows denote ROI areas.

LoDoPaB-CT: (a) GT, (b) FBP, (k) SAD-5, (l) SAD-1 — best PSNR/SSIM with 1 step

Inverse Problem

DPS Chung et al., ICLR 2023

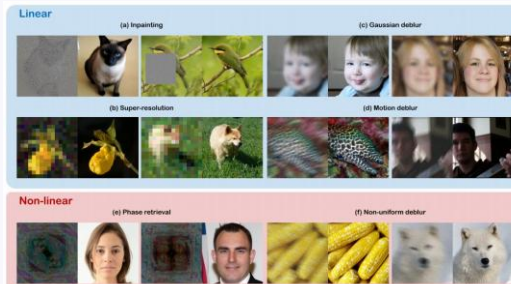


Figure 1: Solving noisy linear and nonlinear inverse problems with diffusion models. Our reconstruction results (right) from the measurements (left) are shown.

Linear (inpainting, SR, deblur) & nonlinear (phase retrieval) inverse problems

Cross-modality

SynDiff Özbey et al., 2023

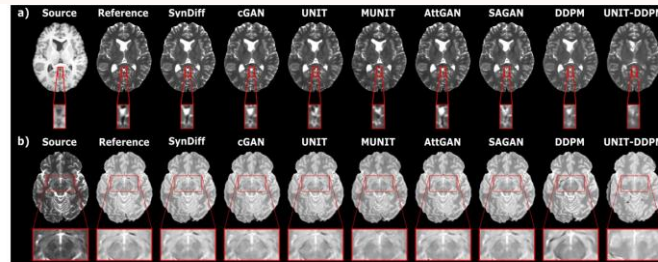


Fig. 3: SynDiff was demonstrated on IXI for translation between MRI contrasts. Synthesized images from competing methods are displayed along with the source and the ground-truth target (reference) images for representative a) $T_1 \rightarrow T_2$, b) $T_2 \rightarrow PD$ tasks. Display windows of a) [0 0.65], b) [0 0.80] are used. Compared to baselines, SynDiff yields lower noise and artifacts, and maintains higher anatomical fidelity.

IXI dataset: (a) $T_1 \rightarrow T_2$, (b) $T_2 \rightarrow PD$ synthesis — lower noise, higher anatomical fidelity

Diffusion & Bridge Models

Image-to-Image Schrödinger Bridge (I²SB)

♦ [Liu et al., ICML 2023]

$$\mu_t = \frac{\bar{\sigma}_t^2}{\bar{\sigma}_t^2 + \sigma_t^2} X_0 + \frac{\sigma_t^2}{\bar{\sigma}_t^2 + \sigma_t^2} X_1$$

- ▶ **Tractable Nonlinear Bridge:** Establishes a direct Schrödinger Bridge between X_{low} and X_{high} , strictly bypassing the pure Gaussian noise state
- ▶ **Analytic Transition:** The intermediate mean(μ_t) is analytically formulated as a time-dependent convex combination of the boundary pairs
- ▶ **Prior Preservation:** Inherently preserves the spatial scaffold of the low-harmonic input throughout the generation trajectory

Residual Denoising Diffusion Models (RDDM)

♦ [Liu et al., CVPR 2024]

$$I_{t-1} = I_t - (\bar{\alpha}_t - \bar{\alpha}_{t-1})I_{res}^\theta - (\bar{\beta}_t - \bar{\beta}_{t-1})\epsilon_\theta$$

- ▶ **Dual Diffusion Process:** Explicitly decouples the conventional reverse diffusion into independent residual and noise pathways
- ▶ **Deterministic Guidance:** The predicted residual(I_{res}^θ) serves as a directional vector, strictly guiding the generation toward X_{high}
- ▶ **Stochastic Perturbation:** The learned noise(ϵ_θ) injects controlled variance to recover high frequency details without over-smoothing

SDE & Flow Models

Image Restoration with Mean-Reverting SDE (IR-SDE)

◆ [Luo et al., ICML 2023]

$$dx = [\theta_t (\mu - x) - \sigma_t^2 \nabla_x \log p_t(x)] dt + \sigma_t d\hat{w}$$

- **Reverse-Time Restoration:** Formulates the generation process as a reverse-time SDE, governed by two complementary driving forces
- **Mean-Reverting Anchor:** The drift term $\theta_t (\mu - x)$ continuously pulls the trajectory toward the low-harmonic input μ , ensuring robust structural preservation
- **Score-Guided Generation:** The learned score function $\nabla_x \log p_t(x)$ systematically estimates the data distribution to recover the missing high-harmonic details

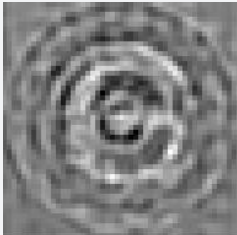
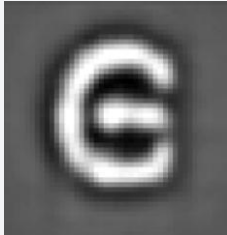
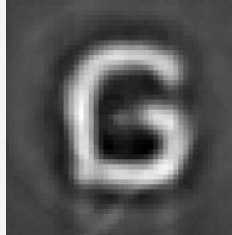
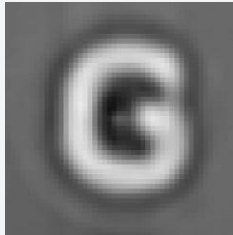
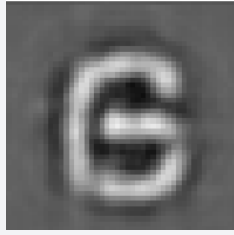
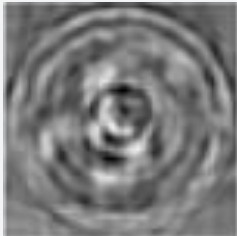
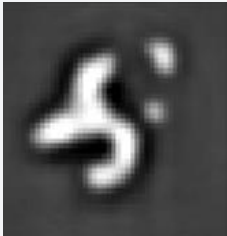
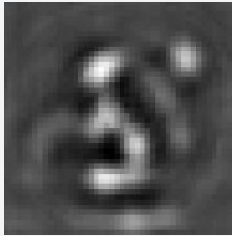
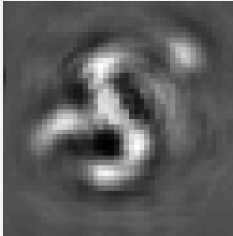
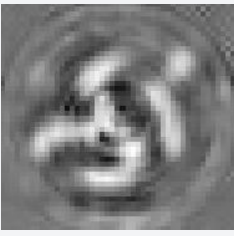
Reversing Flow for Image Restoration (ResFlow)

◆ [Qin et al., CVPR 2025]

$$\frac{\partial [x_t^T; y_t^T]^T}{\partial t} = v_\theta(x_t, y_t, t)$$

- **Deterministic Degradation Flow:** Models the image degradation process as a fully reversible, deterministic path using Continuous Normalizing Flows(CNF)
- **Auxiliary-Augmented State:** Strategically augments the ODE state(x_t) with an auxiliary variable(y_t) to systematically disambiguate the uncertainty scope inherent to ill-posed restoration
- **Velocity-Matched Inversion:** Learns the precise velocity field(v_θ) to uniquely map the combined state back to the high-harmonic target(X_{high}), enabling accurate restoration in fewer than 4 sampling steps

Intra-Harmonic Restoration (5th $\rightarrow$ 5th)

Input (5th harmonic)	I ² SB	RDDM	IR-SDE	ResFlow				
								
								
	SSIM 0.4773	PSNR 15.2700	SSIM 0.4461	PSNR 13.2459	SSIM 0.4430	PSNR 13.5280	SSIM 0.4458	PSNR 13.6397

⚠ **Baseline:** Intra-harmonic (5th $\rightarrow$ 5th) — same frequency band. Severely noise-dominated input causes generalization bottlenecks under zero-shot / OOD conditions.

Why 3rd Harmonic as Input?

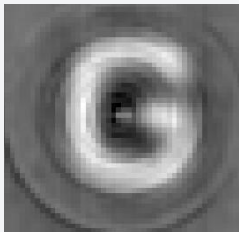
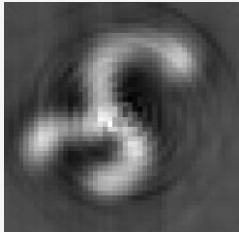
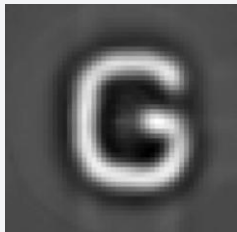
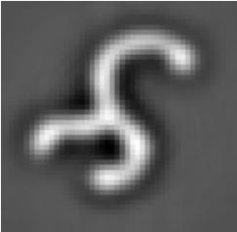
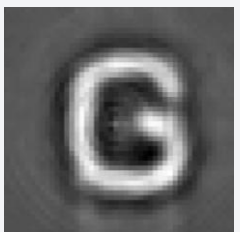
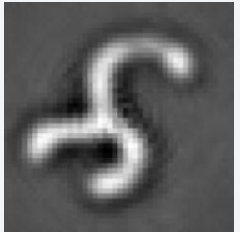
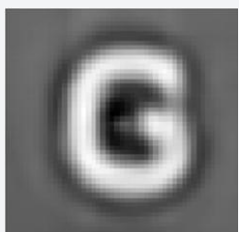
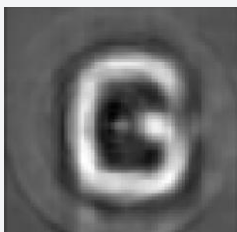
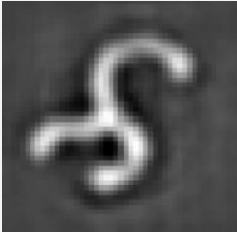
⚠ 5th Harmonic as Input: Limitations

- ▶ **Noise-Dominated Signal:** Low-concentration 5th-harmonic signals decay rapidly, leaving inputs almost entirely dominated by noise with almost no recoverable structure
- ▶ **OOD Generalization Failure:** Under zero-shot conditions with unseen geometries or altered noise, models frequently confuse real structures with system-induced artifacts
- ▶ **Inadequate Diffusion Prior:** Modeling severe concentration-dependent noise as simple Gaussian processes is both theoretically and empirically inadequate for robust MPI recovery

✓ 3rd Harmonic: Cross-Harmonic Rationale

- ▶ **High-SNR Geometric Scaffold:** The 3rd harmonic maintains high SNR even at low concentrations, providing a reliable and stable spatial draft for the generative process
- ▶ **Noise-Free Generative Prior:** Bypasses the need to model complex concentration-dependent noise dynamics; the network resolves high-frequency detail from a plausible low-frequency prior
- ▶ **Robust OOD Generalization:** Even for novel particle distributions, the model recovers high-frequency details from a reliable geometric scaffold rather than synthesizing structure from pure noise

Proposed Cross-Harmonic Strategy (3rd $\rightarrow$ 5th)

Input (3rd harmonic)	I ² SB	RDDM	IR-SDE	ResFlow				
 	 	 	 	 				
	SSIM 0.6660	PSNR 16.8939	SSIM 0.6689	PSNR 17.9238	SSIM 0.6047	PSNR 16.1892	SSIM 0.6329	PSNR 16.9683

◆ **Key Result:** Cross-harmonic achieves **+0.19 SSIM** · **+1.6 dB PSNR** vs. intra-harmonic baseline. 3rd-harmonic scaffold prevents arbitrary structure generation and ensures robust OOD generalization.

Limitations & Impacts

01

Paired Training Data

Generative denoising methods require (LQ, HQ) paired training data. Such paired acquisitions are costly and difficult to obtain in MPI.

► Impact

Acquiring matched LQ-HQ MPI phantom pairs demands precise experimental conditions and significant hardware resources, limiting scalability.

02

Dual-Concentration Setup

Our generative framework requires perfectly aligned paired data across different tracer concentrations, increasing logistical complexity compared to conventional single-scan methods

► Impact

Maintaining absolute spatial consistency between low- and high-concentration scans requires careful calibration, limiting clinical or multi-site scalability.

03

Gaussian Noise Assumption

Diffusion-based methods define degradation as cumulative Gaussian noise. MPI measurement noise, however, cannot be fully modeled by this assumption.

► Impact

MPI system noise is concentration-dependent and structured, making a pure Gaussian diffusion prior a theoretically incomplete model for this inverse problem.